SUMULONG MEMORIAL HIGH SCHOOL
Gen. Luna St. Brgy. San Roque, Antipolo City
SY. 2022-2023

LIGHT AND SOUND

LEARNING MODULE

NAME: ______________________________ **YEAR AND SECTION:** __________________

SET THE TARGET

LEARNING OBJECTIVES:

At the end of this module, you are expected to:
1. Infer how the movement of particles of an object affects the speed of sound;
2. Explain the hierarchy of colors in relation to the energy of visible light.
3. Differentiate between heat and temperature at the molecular level

MOTIVATIONAL ACTIVITY

REBUS! Direction: Identify the picture, followed by mathematical equation to reveal the answer.

1. H +  =

2.  + W +  - P + V =

3.  +  - N =

4. L +  - S =

5.  - E + ION =

Overview:

In grade 7, you discussed that energy is transferred or transmitted from one object to another. When sound is created the wave (sound) is transmitted by air to vibrate transferring the energy.

The Science of Sound has gone all the way from a mere transfer of energy to the creation of tunes and music for entertainment. Most of our gadgets are sound embedded to amuse us. The newest focus of sound science is on ecology where ecological patterns and phenomena are predicted based on sounds released by the different components of the ecosystem.

The Science of Light has gone all the way from a mere transfer of energy to the creation of colors for entertainment and other purposes. Most of our gadgets are light emitting for efficiency when used at night. But the most important purpose is for humans and other animals to see the beautiful world through light.

So, are you ready to have fun with sounds and explore the characteristics and properties of light?

- **SOUND**
  - Exhibits wave properties that produced by vibrating sources.
  - It cannot travel through a vacuum and it is capable of travelling through different types of the medium (solid, gas and liquid).
- **LIGHT**
  - A form of wave.
  - When the light strikes an object, the light can be reflected, absorbed or transmitted.

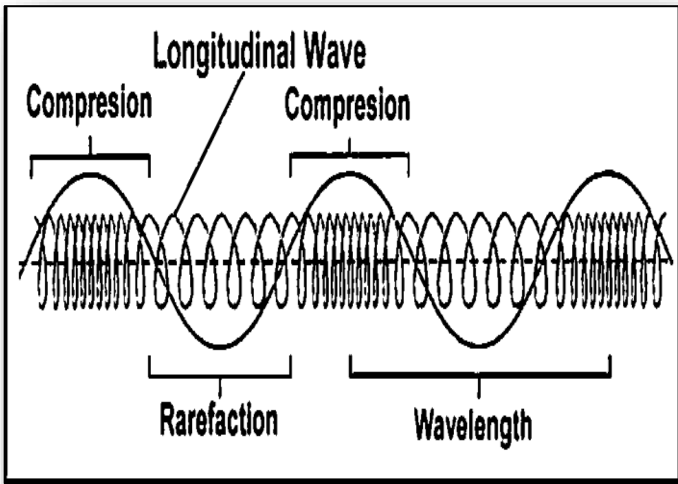
SOUND PROPAGATION

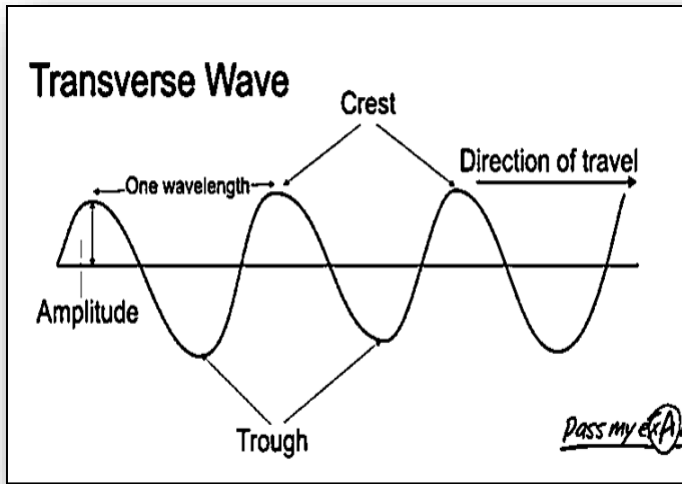
Sound consists of waves of air particles. Generally, sound propagates and travels through air. It can also be propagated through other media. Since it needs a medium to propagate, it is considered a mechanical wave. In propagating sound, the waves are characterized as:

LONGITUDINAL WAVE. – These are waves that travel parallel to the motion of the particles.

TRANSVERSE WAVE – These are waves that travel perpendicular to the direction of vibrations

The compressions resemble the trough while the rarefactions are the crests.



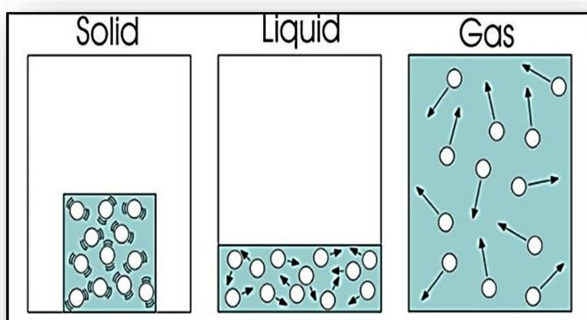


CHARACTERISTICS OF WAVES

- **TROUGH** is the lowest part of a transverse wave
- **CREST** is the highest portion.
- **WAVELENGTH (λ)**The distance from one compression to the next or between two successive compressions in a longitudinal wave equals.
- **FREQUENCY(f)** The number of vibrations in 1 second. The SI Unit is the HERTZ (Hz). (1 Hz=1 sec, 50 Hz=1sec)
- **SPEED (v)** The product of wavelength and the frequency
- **AMPLITUDE (A)** The height of Crest or depth of a Trough measured from the normal undisturbed position.
- **PERIOD (T)** the time taken from one complete vibration

  - To get the Speed of the wave, the equation is: $\mathbf{v=f\lambda}$

Sound waves travel at different velocity depending on the medium and how closely packed the molecules in the matter. Of the three media (solid, liquid and gas), sound travels fastest through SOLIDS, faster through LIQUID and slowest through GASES. This can be attributed to the properties of the medium such as elasticity, density, and temperature.



- **ELASTICITY –** the ability of the material to bounce back after being disturbed. A more elastic medium allows sound to travel quickly because when the particles are compressed, they quickly spread out again.

- **DENSITY –** depends on how much matter there is in a given amount of space. Sound travels more slowly in denser materials since the denser the medium, the more mass it has given volume. The particles of a dense material do not , move as quickly as those of the less-dense medium.

- **TEMPERATURE –** a factor that affects the speed of sound. In medium, sound travels more slowly at lower temperature and faster at higher temperature.

Sound travels at about 331 m/s in dry air at 0°C. The speed of sound is dependent on temperature of the medium where an increase is observed with an increase in temperature. This means that at temperatures greater than 0oC speed of sound is greater than 331m/s by an amount 0.6 m/s of the temperature of the medium. In equation,

$$\boldsymbol{v = 331\frac{m}{s} + 0.6\frac{\frac{m}{s}}{°C}(T)}$$

where T is the temperature of air in Celsius degree and 0.6 m/s /°C is a constant factor of temperature. Let's try it out at a room temperature of 25 Celsius.

SAMPLE PROBLEM

PROBLEM:
What is the speed of sound in air of temperature 25°Celsius?

SOLUTION:
Given: T= 25°C

$$\boldsymbol{v = 331\frac{m}{s} + 0.6\frac{\frac{m}{s}}{°C}(T)}$$

$$\boldsymbol{v = 331\frac{m}{s} + 0.6\frac{\frac{m}{s}}{\cancel{°C}}(25)\cancel{°C}}$$

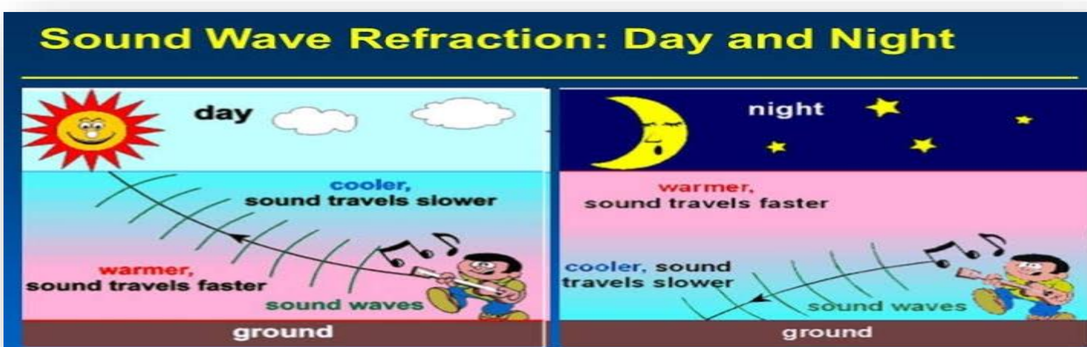
$$\boldsymbol{v = 331\frac{m}{s} + 15\frac{m}{s}}$$

$$\boxed{\boldsymbol{v = 346\frac{m}{s}}}$$

PROPERTIES OF WAVES

- **REFLECTION -** described as the turning back of a wave as it hits a barrier.
  Echo is an example of a *reflected sound*. **Reverberation** on the other hand refers to the *multiple reflections or echoes in a certain place*. A reverberation often occurs in a small room with height, width, and length dimensions of approximately 17 meters or less.

- **REFRACTION -** described as the change in speed of sound when it encounters a medium of different density. Sound is heard better in far areas during nighttime than during daytime. This change in speed of sound during refraction is also manifested as sort of "bending" of sound waves.



Do all these terms and concepts seem confusing? Let's try the succeeding activities to get a clearer picture of what sound waves and its transmissions are.

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ACTIVITY 1: THE DANCING SALT

OBJECTIVE:

At the end of the activity, you will be able to

- Infer that sound consists of vibrations that travel through the air

MATERIALS:

1 rubber band
1 piece of plastic sheet
1 empty large can of powdered milk - 800 g
1 wooden ruler
1 empty small can of evaporated milk - 400 mL
rock salt
transistor radio

PROCEDURES:

Vibrations produce sound

1. Prepare all the materials needed for the activity. Make sure that you find a work where you can independently do the activity.

2. Put the plastic tightly over the open end of the large can and hold it.Puts the rubber band over it.

3. Sprinkle some rock salt on top of the plastic.

4. Hold the small can close to the salt and tap the side of the small can with the ruler as shown in Figure 2

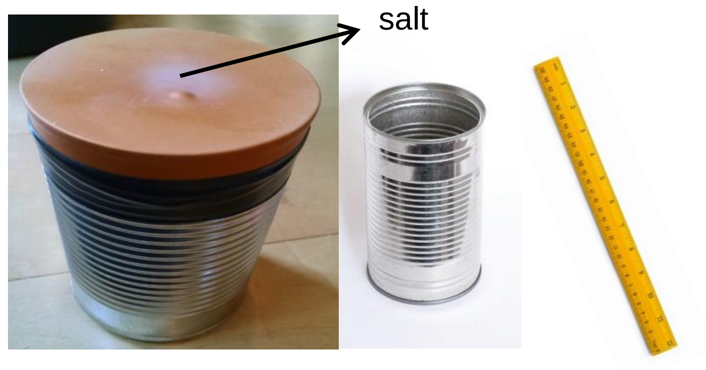

Figure 2

Q1. What happens to the salt?

5. Try tapping the small can in different spots or holding it in different directions. Find out how you should hold and tap the can to get the salt to move and dance the most.

Q2. How were you able to make the salt move and dance the most?

Q3. What was produced when you tapped the small can? Did you observe the salt bounce or dance on top of the plastic while you tapped the small can?

Q4. What made the salt bounce up and down?

Q5. From your observations, how would you define sound?

6. Switch on the transistor radio and position the speaker near the large can. Observe the rock salt.

7. Increase the volume of the radio while it is still positioned near the large can.
Observe the rock salt again.

Q6. What happened to the rock salt as the loudness is increased?

Q7. Which wave characteristic is affected by the loudness or the intensity of sound?

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ACTIVITY 2: SOUND RACE…WHERE DOES SOUND TRAVELS FASTEST?

OBJECTIVE:

At the end of the activity, you will be able to:

- Distinguish which material transmits sound the best.

MATERIALS:

watch/clock that ticks
mobile phone
wooden dowel 80-100 cm long
metal rod 80-100 cm long
string (1 meter)
metal spoon
3 pieces zip lock bag (3x3) or waterproof mobile phone carrying case

PROCEDURES:

1. Hold a ticking watch/clock as far away from your body as you can. Observe whether or not you can hear the ticking.
2. Press one end of the wooden dowel against the back part of the watch and the other end beside your ear. Listen very well to the ticking sound. Record your observations.
3. Repeat step #2 using a metal rod instead of the wooden dowel. Record your observations.

Q1. Did you hear the watch tick when you held it at arm's length? When you held it against the wooden dowel? When you held it against the metal rod?

4. Repeat steps #1 to #3 using a vibrating mobile phone instead. Record your observations.

Q2. Did you hear the mobile phone vibrate when you held it at arm's length? When you held it against the wooden dowel? When you held it against the metal rod?

5. Place the mobile phone in the waterproof carrying case and dip it in a basin of water while it vibrates.

Q3. Based on your observations, which is a better carrier of sound? Air or wood? Air or water? Air or metal? Water or metal?

6. At the center of the meter long string, tie the handle of the metal spoon. Hold the string at each end and knock the spoon against the table to make it ring or to create a sound. Listen to the ringing sound for a few seconds then press the ends of the strings against your ears. Observe and record the difference in sound with and without the string pressed against your ear.
7. Knock the spoon against the table. When you can no longer hear the sound of the ringing spoon, press the ends of the string against your ears. Record whether or not you could hear the ringing of the spoon again.

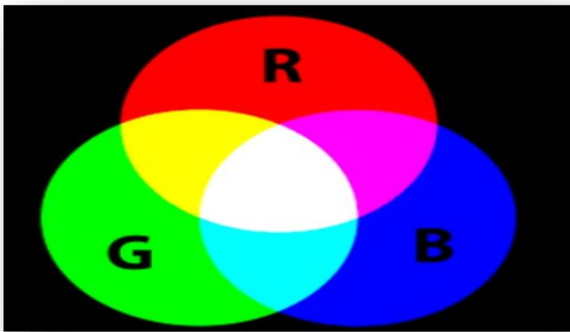
Q4. How did the sound of the spoon change when the string was held against your ears?

Q5. When the ringing of the spoon was too quiet to be heard through the air, could it be heard through the string?

Q6. Is the string a better carrier of sound than air?

LEARN MORE!

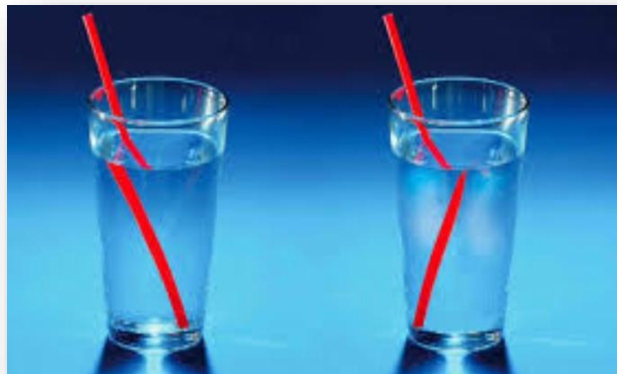
COLORS OF LIGHT



LIGHT is form of energy that can be detected by the human eye.

COLOR is a function of the human visual system. RED, GREEN and BLUE are the primary colors of light- they can be combined in different proportions to make all other colors.

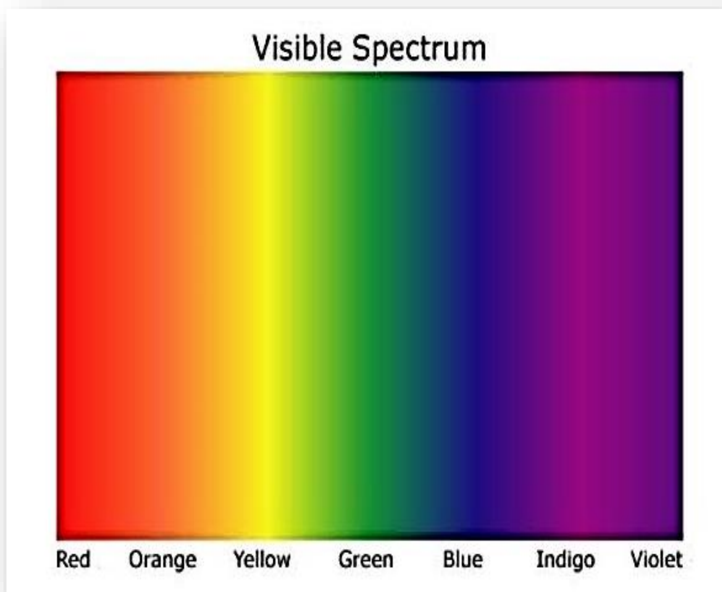
REFRACTION OF LIGHT



It is the change in direction of a wave passing from one medium to another or form gradual change in the medium. It is also the bending of light as it passes from one transparent substance into another. This happens because of the change in speed and orientation of the light with respect to the normal as it traverses a new medium of a different density.

VISIBLE LIGHT SPECTRUM

If light waves are passed through a prism, it is refracted into a rainbow of colors. Each color has a different wavelength, longer wavelengths refract less than shorter wavelengths. The visible spectrum is the range of wavelengths of electromagnetic radiation that is normally visible from 400-700 nanometers (nm). White light is a mixture of colors that conventionally divided into seven (7) major colors: RED, ORANGE, YELLOW, GREEN, BLUE, INDIGO, and VIOLET (ROYGBIV). Red has the smallest refraction and Violet has the greatest.



PRODUCING VISIBLE LIGHT

- INCANDESCENT LIGHTS
- FLOURESCENT LIGHTS
- NEON LIGHTS
- SODIUM VAPOR LIGHTS
- TUNGSTEN LIGHTS
- BIOLUMINISCENCE

PRIMARY COLORS

- Red, Green and Blue

SECONDARY COLORS

- Cyan, Yellow, and Magenta

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ACTIVITY 3: HOW DOES LIGHT TRAVEL?

OBJECTIVES:

1. To investigate how light travels

MATERIALS:

A candle, matches, and three sheets of paper.

PRE LABORATORY QUESTION:

- How does one see light?

PROCEDURES:

1. Punch a small hole in the middle of each of the three sheets of paper.
2. Look at a lighted candle through the hole in the first sheet of paper.
3. Place the second sheet of paper on the top (but without touching) the first paper again observe the candle through the hole.
4. Do the same with the third sheet of paper. Make sure that the sheets of paper are not in contact with each other. What have you notice about the hole in the paper

CONCLUSIONS:

From this activity, what can you conclude about how light travels?

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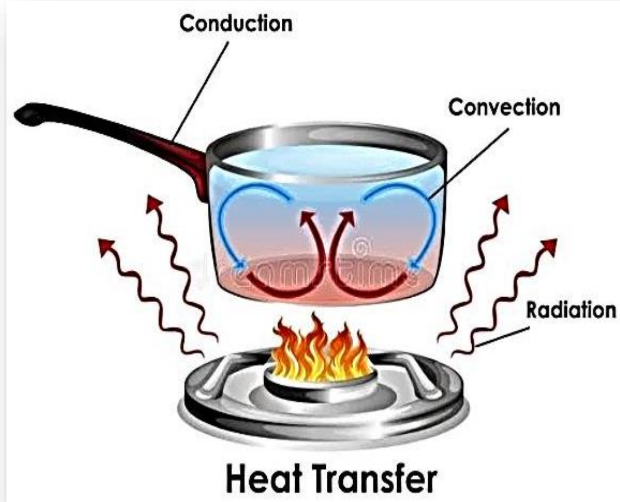
LEARN MORE!

HEAT AND TEMPERATURE

Heat transfer happens around us all the time. Although we do not see how this process actually takes place, its effects are evident. In fact, we rely on these effects every day in many of the activities that we do. Understanding the concepts behind heat transfer therefore helps us do our activities more efficiently.

TEMPERATURE AND HEAT are two different concepts but they are related to each other.

HEAT is the transfer of energy between objects or places because of difference in temperature. Heat exists as 'energy in transit' and it is not contained in an object. The energy that is actually contained in an object due to the motion of its particles is called thermal energy. The thermal energy of an object is changed if heat is transferred to or from it.

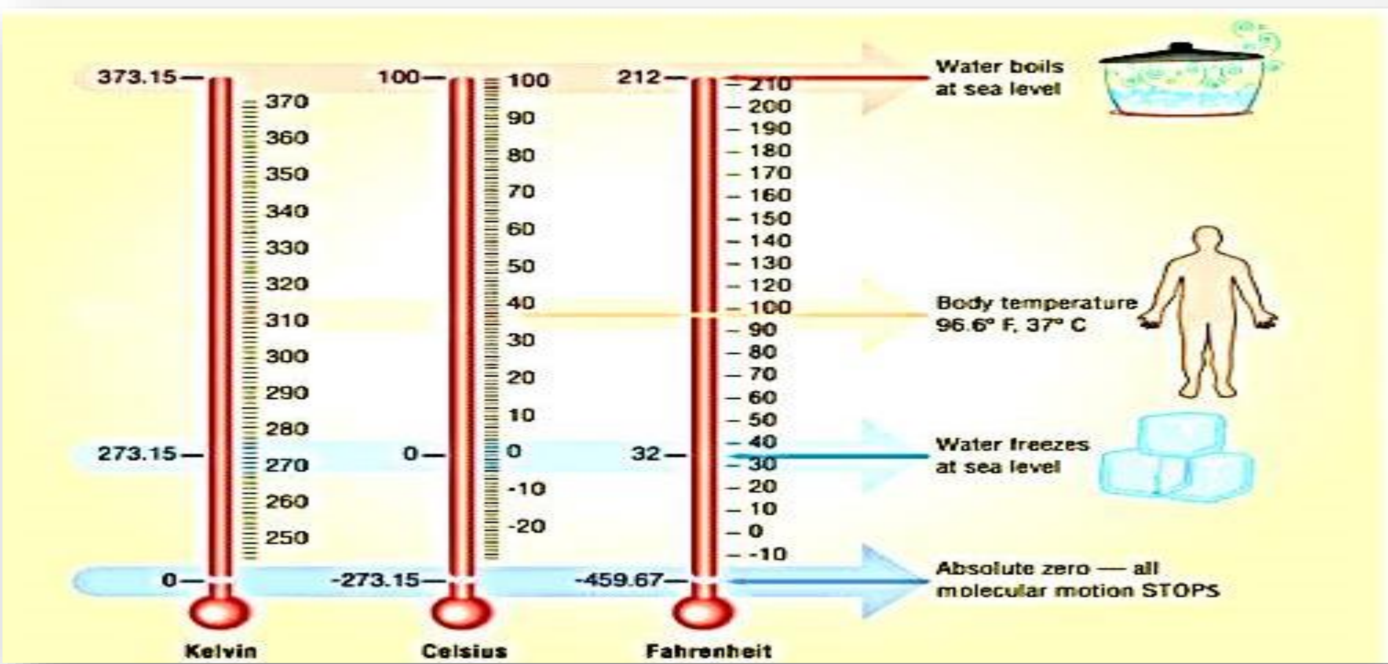


TYPES OF HEAT TRANSFER

- **CONDUCTION –** takes place between solids or solids with liquids. Particles vibrating or moving faster transfer some of their energy to nearby atoms.
- **CONVECTION –** occurs in fluids (liquids and gases). Caused by a change in density due to a change in temperature.
- **RADIATION –** transfer of energy due to infrared and other electromagnetic rays. Radiation can travels through space.

TEMPERATURE is the measure of the degree of hotness or coldness of the body. It is also related to the random motion of the molecules in a substance. In relation to heat, it is defined as the property that determines the transfer of heat energy to or from other objects. THERMOMETER is the instrument used in measuring temperature.

TEMPERATURE SCALES

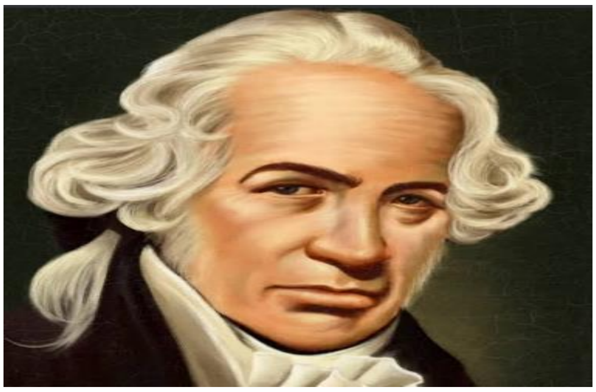


ANDERS CELSIUS
(1701- 1744)

A Swedish astronomer, shown the importance of pure water. The CELSIUS scale is used for common temperature measurements in most of the world. 0°C is the zero point being defined by the freezing point of water and 100°C was the boiling water both at sea level atmospheric pressure.

FORMULA:

$$°C = (°F - 32) \times \frac{5}{9}$$

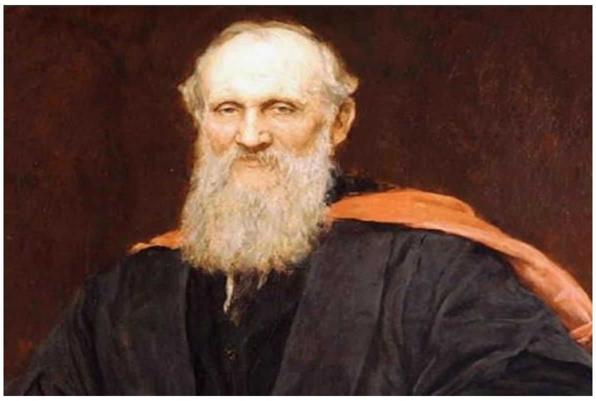


DANIEL GABRIEL FAHRENHEIT
(1685-1736)

The first to create a scale among three. His basis was the freezing and boiling point of water. FAHRENHEIT commonly used in USA, which water freezes at 32°F at 212 °F at sea level atmospheric pressure.

FORMULA:

$$°F = °C \times \frac{9}{5} + 32$$



BARON KELVIN
(1824 – 1907)

Introduced the concept of the absolute zero, the lowest possible temperature. KELVIN is the instrument used in many scientific measurements. The freezing point is 273.16° K at sea level atmospheric pressures.

FORMULA:

$$°K = °C + 273.16$$

SAMPLE PROBLEMS:

1. Convert 40°C to Fahrenheit scale

FORMULA: $°F = °C \times \frac{9}{5} + 32$

SOLUTION:

$°F = 40° \times \frac{9}{5} + 32$

$°F = \frac{360°}{5} + 32$

$°F = 72° + 32$

$\boxed{°F = 104}$

2. Convert 86°F to Celsius scale

FORMULA: $°C = (°F - 32) \times \frac{5}{9}$

SOLUTION:

$°C = (86° - 32) \times \frac{5}{9}$

$°C = (86° - 32) \times \frac{5}{9}$

$°C = 54° \times \frac{5}{9}$

$°C = \frac{270°}{9}$

$\boxed{°C = 30}$

3. Liquid oxygen boils at normal pressure at -182.96 °C. What is this temperature in Kelvin?

SOLUTION:

$K = °C + 273.15$

$K = -182.96 + 273.15$

$\boxed{K = 90.19\ K}$

4. The average body temperature of a person is 98.6 °F. What is the average body temperature of a person in Kelvin?

SOLUTION: Plug 96.8 into °F in the conversion factor above.

$K = \frac{5}{9}(°F – 32) + 273.15$

$K = \frac{5}{9}(98.6 – 32) + 273.15$

$K = \frac{5}{9}(66.6) + 273.15$

$K = 37 + 273.15$

$\boxed{K = 310.15\ K}$

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ACTIVITY 4: DYE IN WATER

OBJECTIVES:

At the end of this activity, you should be able to explain the scattering of the dye in water at different temperatures.

MATERIALS

3 transparent containers
1 thermometer
3 plastic droppers
hot water
tap water (room temperature)
cold water
dye (Food color)

PROCEDURES

1. Fill the three containers separately with cold water, tap water, and hot water.

2. Measure the temperature of the water in each container. Record your measurements in Table 2 below.

Table 2: Data for Activity 2

Container	Temperature (°C)	Observations
Container 1		
Container 2		
Container 3		

3. With the dropper, place a drop of dye into the center of each container as shown in Figure 2. (Note: It is better if you place drops of dye into the three samples simultaneously.)

4. Carefully observe and compare the behavior of the dye in the three containers. Write down your observations in Table 2.

Q1. What similarities and differences did you observe when a drop of dye was added to each container?

____

____

Q2. In which container did the dye scatter the fastest? In which di it scatter the slowest?

____

____

Q3. How do you relate the temperature of the water to the rate of scattering of the dye?

____

____

Concept Micro

Sound waves exist as variations of pressure in a medium such as air. They travel in this medium in the same way water propagates through water. Sound waves are longitudinal and transverse waves. The speed of sound in air is constant no matter what the speed of the sources is.

Visible lights arranged from the longest wavelength, (ROYGBIV) can be found at the next level of spectrum.

Color is a property of material that interests and pleases everyone. Color consists of the characteristics of light other than spatial and temporal inhomogeneties.

Heat is the thermal energy transferred from one object toanother due to temperature difference. Temperature is the degree of hotness and coldness of the body.

TEST YOURSELF!

TEST I. <u>MATCHING TYPE.</u> Match column A with the correct answer to column B. Write only the letter of the space provided.

COLUMN A

_____1. Visible Light

_____ 2.Heat

_____3. Sound

_____4. Light

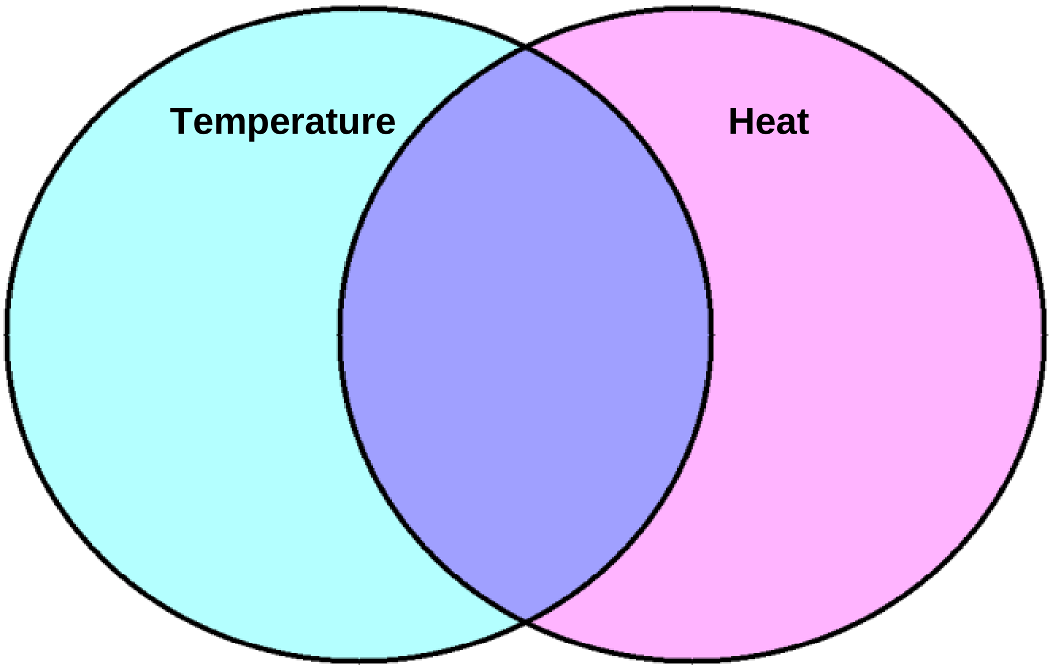
_____5. Color

COLUMN B

A. form of energy that can be detected by the human eye.
B. Exhibits wave properties that produced by vibrating sources
C. the transfer of energy between objects or places because of difference in temperature.
D. factor that affects the speed of sound.
E. function of the human visual system.
F. arranged from the longest wavelength, (ROYGBIV) can be found at the next level of spectrum.

TEST YOURSELF!

TEST II. VENN DIAGRAM. Write details that indicate differences between the two concepts in the outer circles. Write the details that indicate their similarities in the area where the circles overlap.



TEST III. Find the answer in the following problems.

1. Dry ice, or frozen carbon dioxide sublimes (phase change between solid to gas) at -78.5 °C under normal atmospheric pressures. What is this temperature in Fahrenheit?

2. The average body temperature of a house cat is 101.5 °F. What is this temperature in Celsius?

3. Nitrogen is a liquid around 70 K. What is this temperature in Fahrenheit?

TEST IV. ESSAY. Answer briefly the following questions.

1. What is the difference between longitudinal waves to transverse waves?

   ____

   ____

2. Why it is important to control the body temperature?

   ____

   ____

GLOSSARY OF TERMS

COLOR is a function of the human visual system.

HEAT is the transfer of energy between objects or places because of difference in temperature.

LIGHT a form of wave that can reflected, absorbed and transmitted.

LONGITUDINAL WAVE. These are waves that travel parallel to the motion of the particles.

SOUND It exhibits wave properties that produced by vibrating sources.

TRANSVERSE WAVE These are waves that travel perpendicular to the direction of vibrations.

TEMPERATURE is the measure of the degree of hotness or coldness of the body.

VISIBLE SPECTRUM is the range of wavelengths of electromagnetic radiation that is normally visible from 400-700 nanometers (nm).

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